Filter Wizard

Sallen-Key filter

Created on 03/30/2026

ANALOG DEVICES

Filter Wizard Design Report

Filter Requirements for High-Pass, 4th order Butterworth

Specifications: Optimize: Noise; +Vs: 5; -Vs: -5

Gain: 0 dB

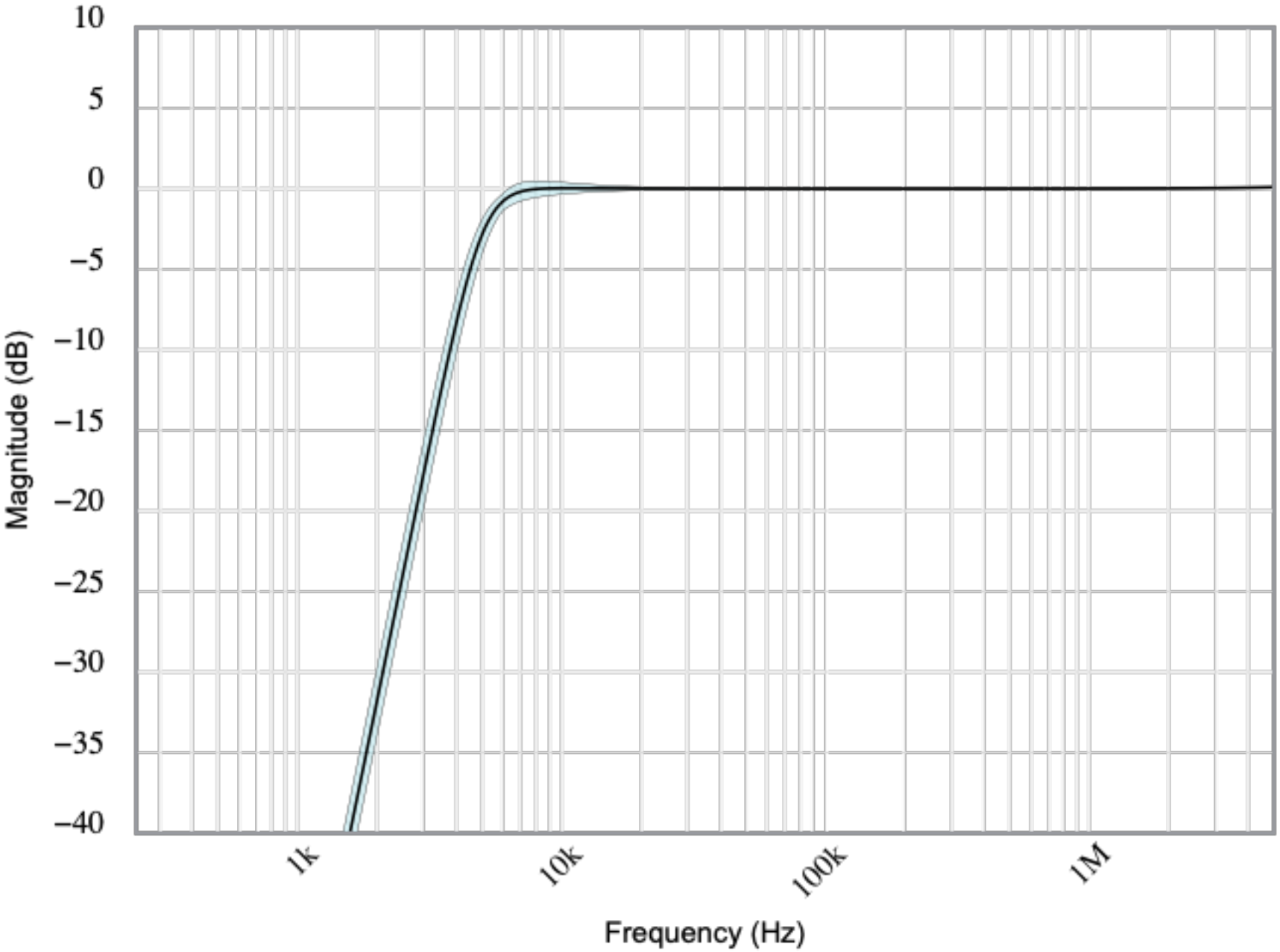
Passband: -3dB at 5kHz

Stopband: -20dB at 2.5kHz

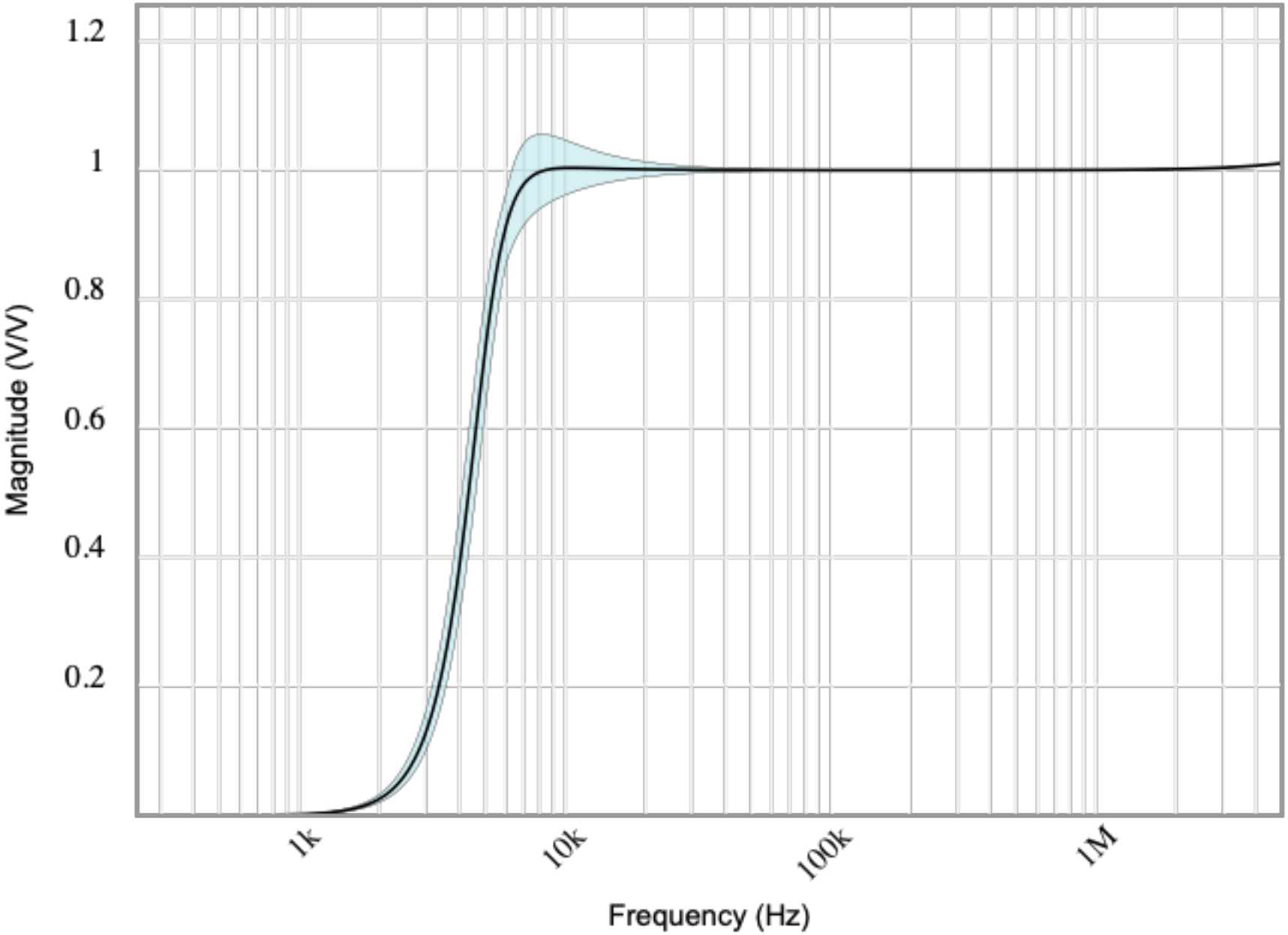
Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

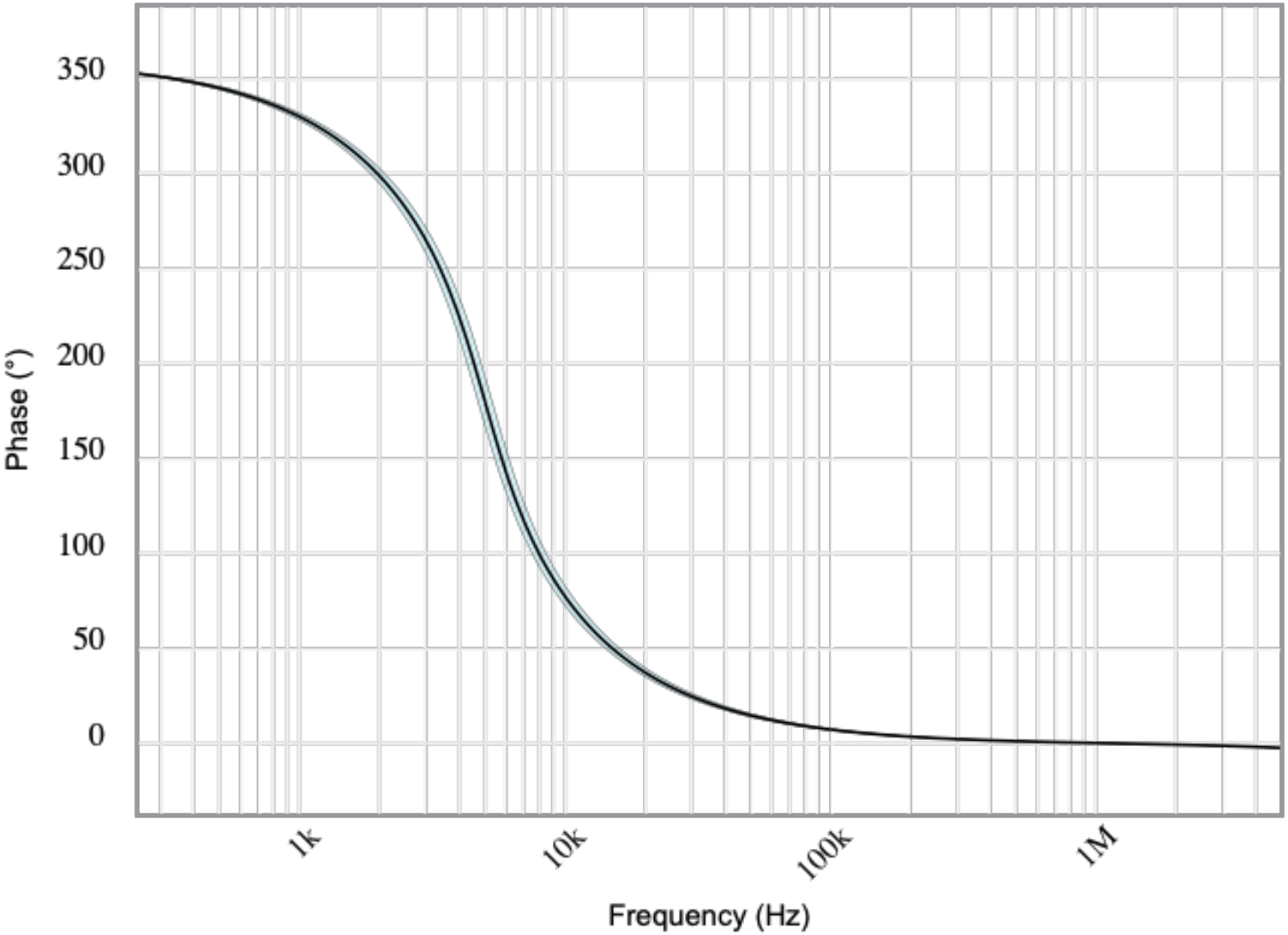
Magnitude(dB)



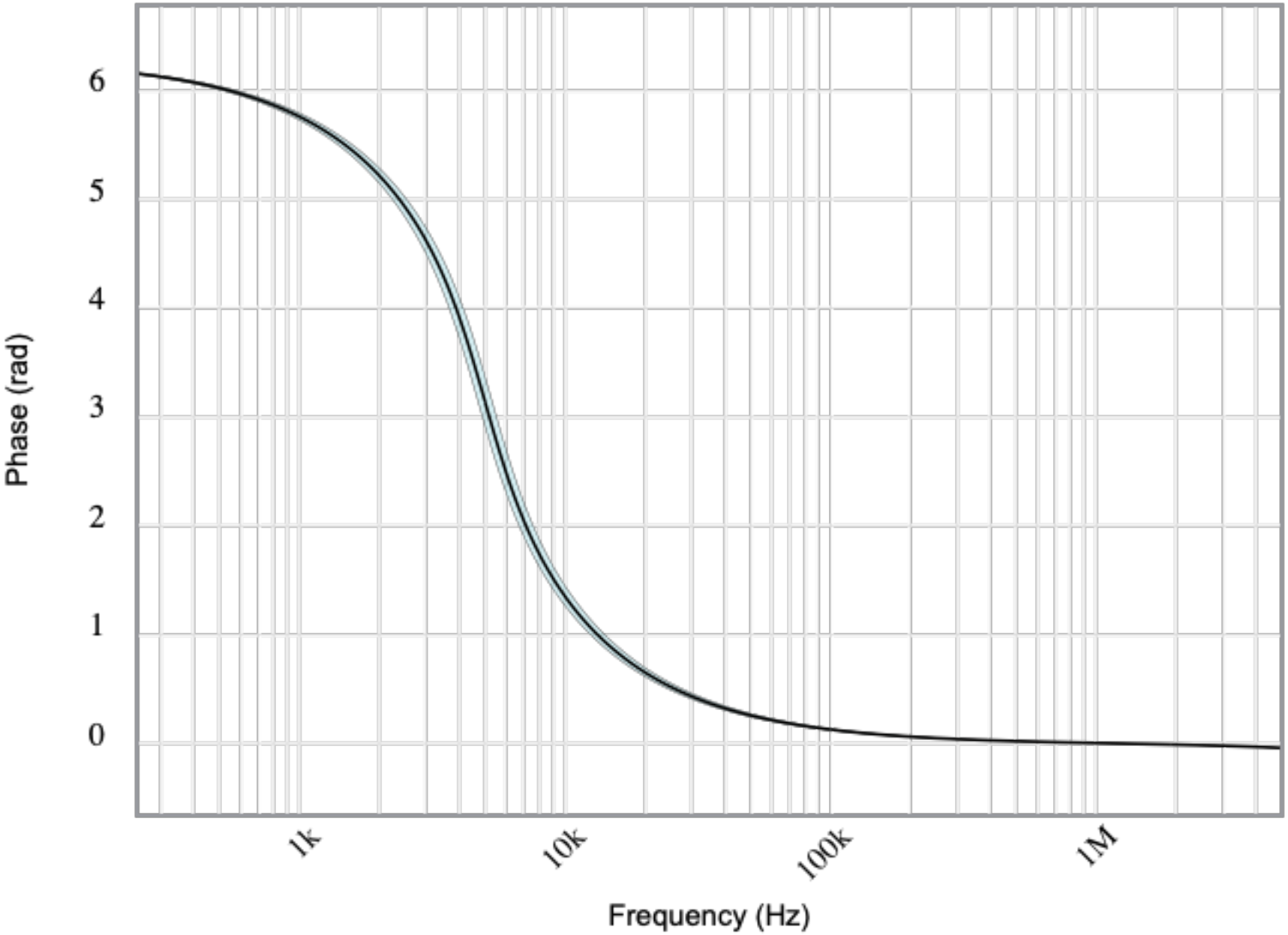
Magnitude(Volts per Volt)



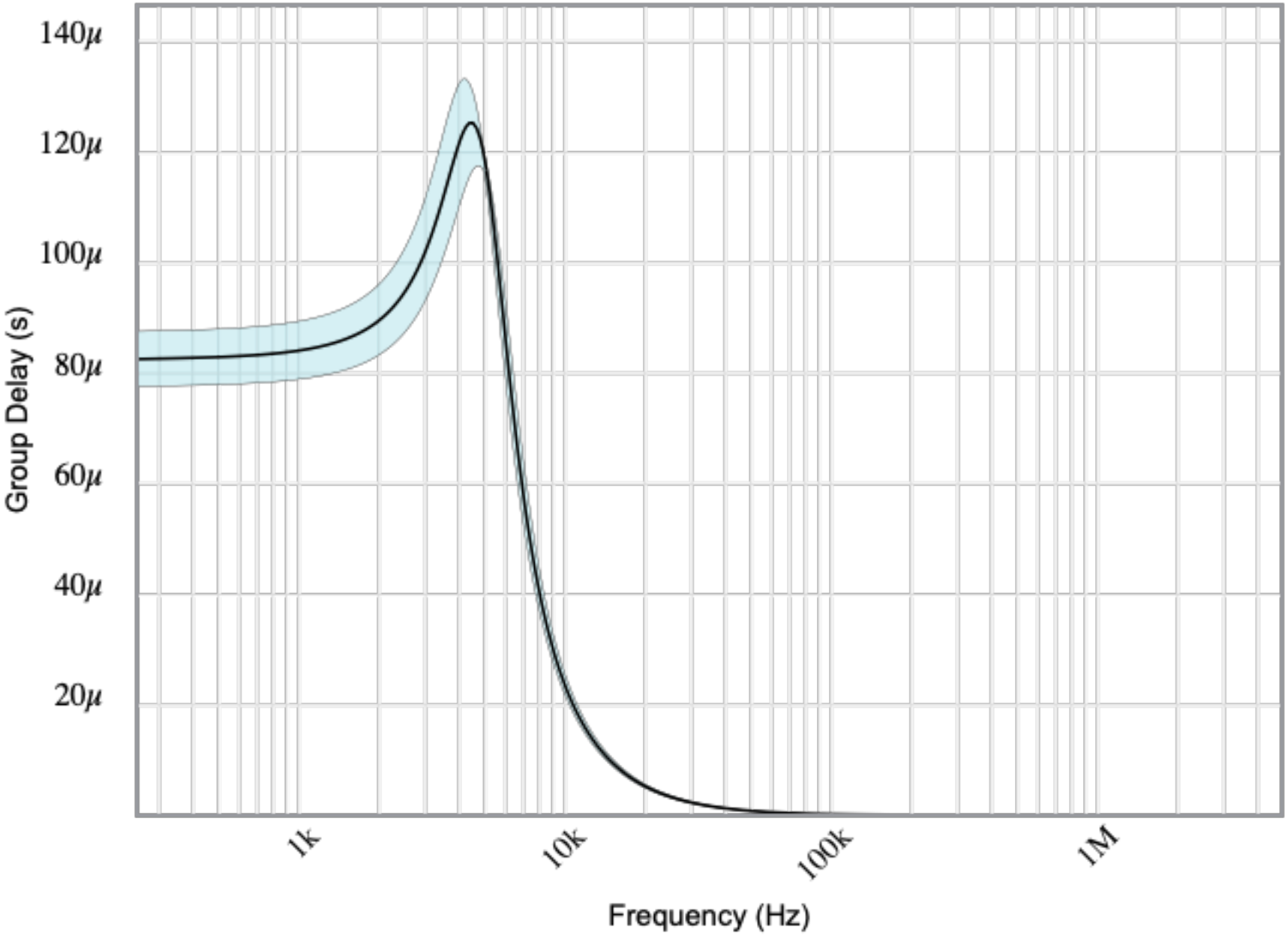
Phase(degrees)



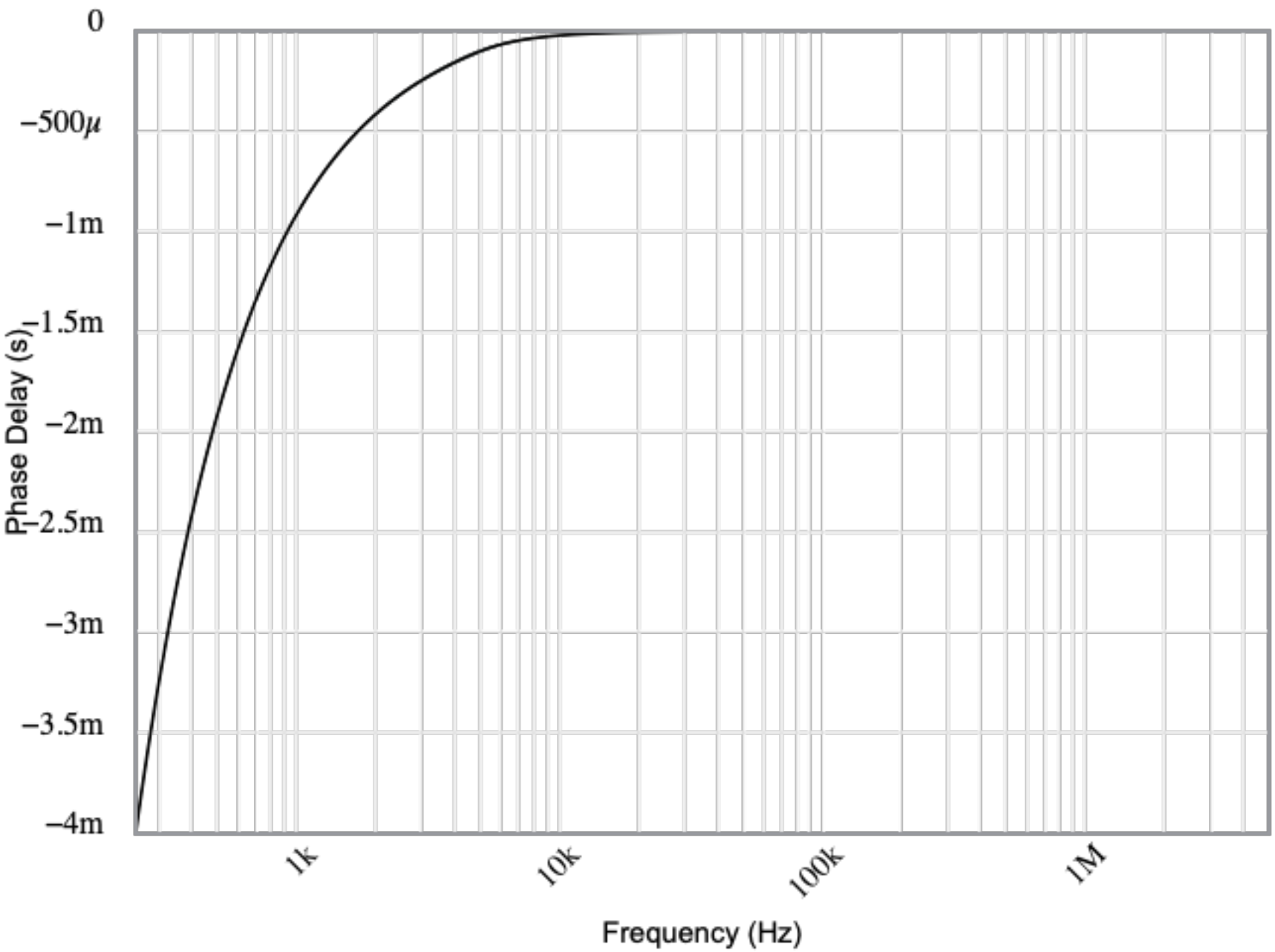
Phase(radians)



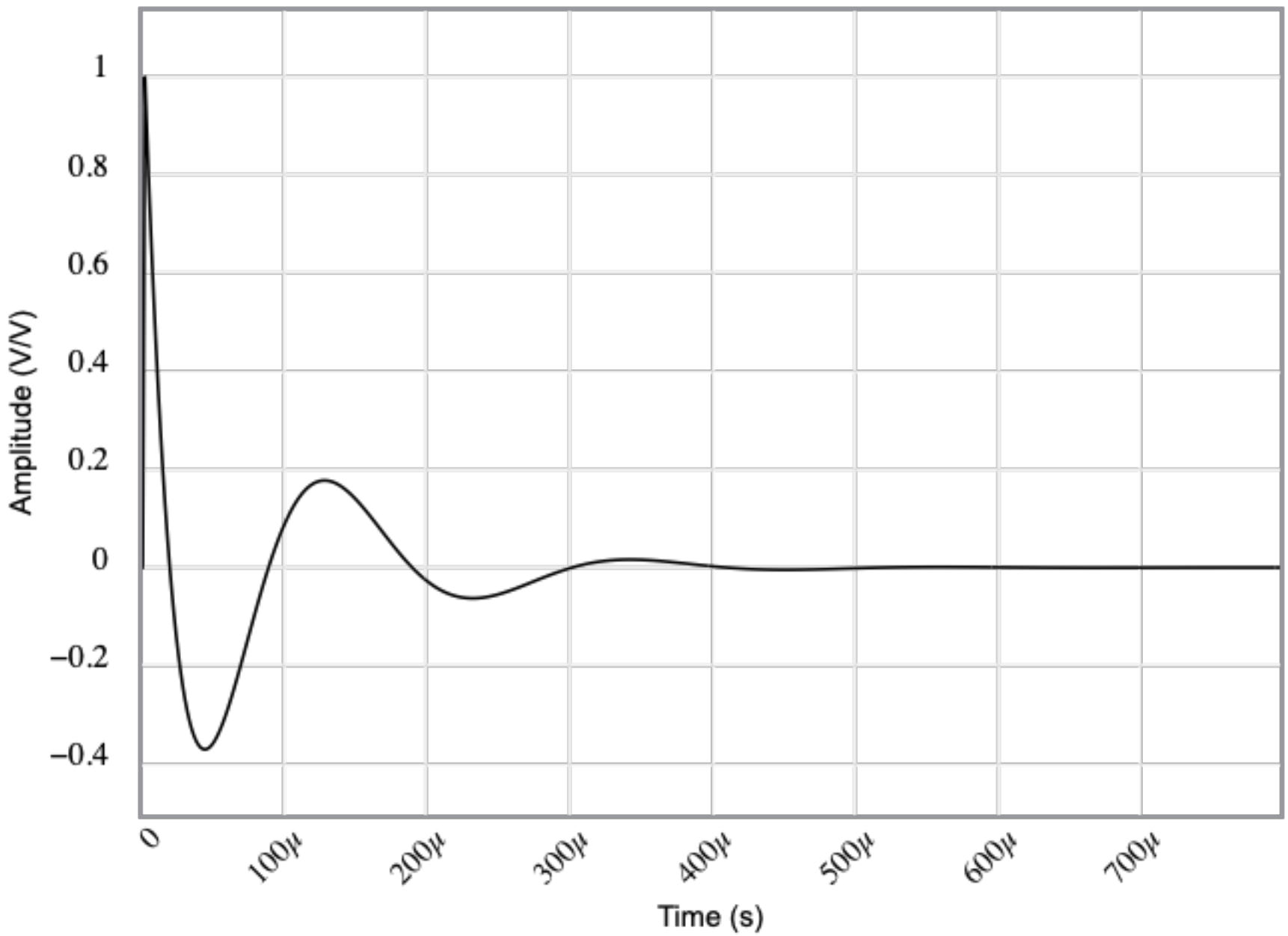
Group Delay


140μ
120μ
100μ
80μ
60μ
40μ
20μ
Group Delay (s)
1k
10k
100k
1M
Frequency (Hz)

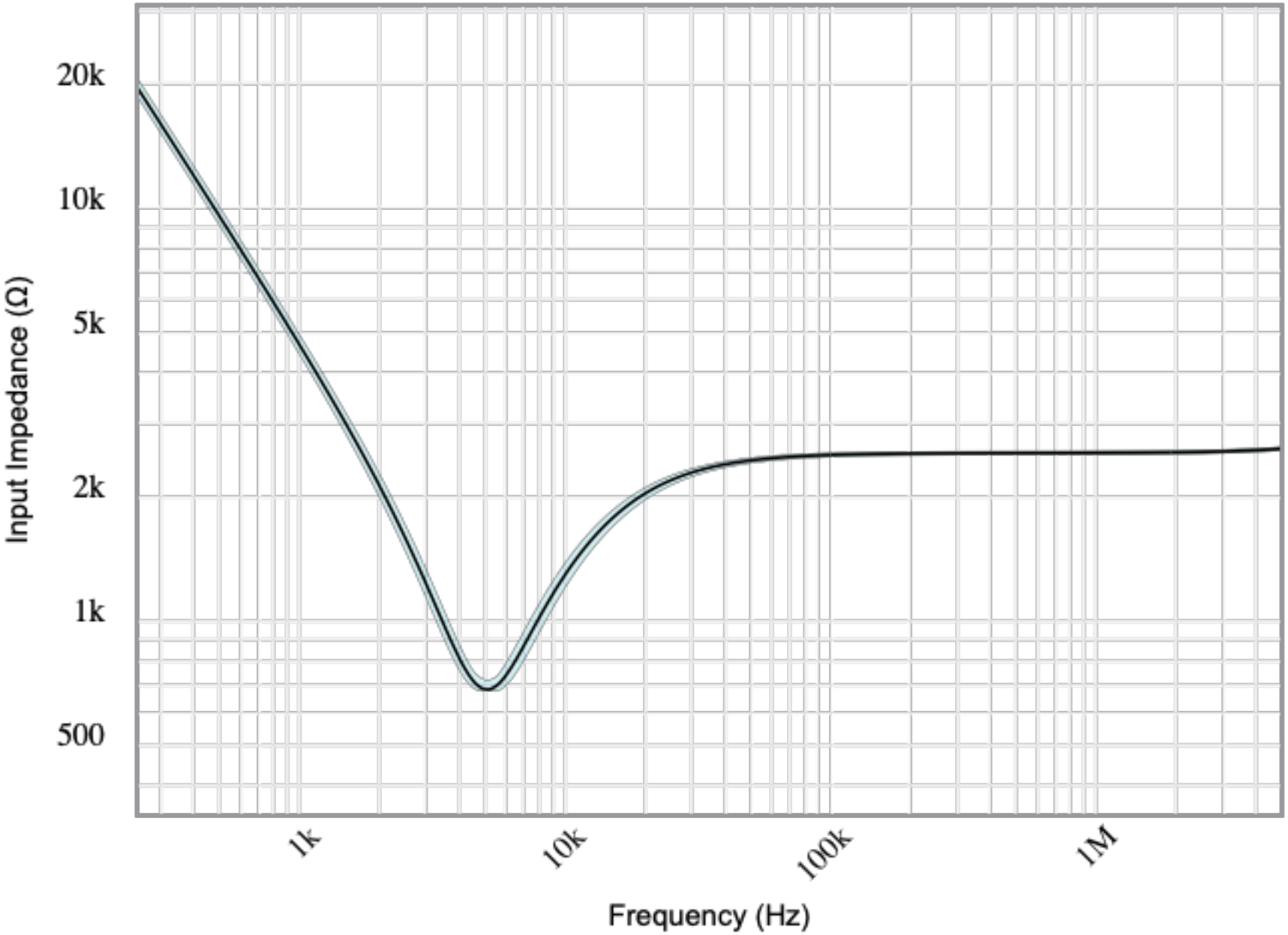
Phase Delay


0
−500μ
−1m
−1.5m
−2m
−2.5m
−3m
−3.5m
−4m
Phase Delay (s)
1k
10k
100k
1M
Frequency (Hz)

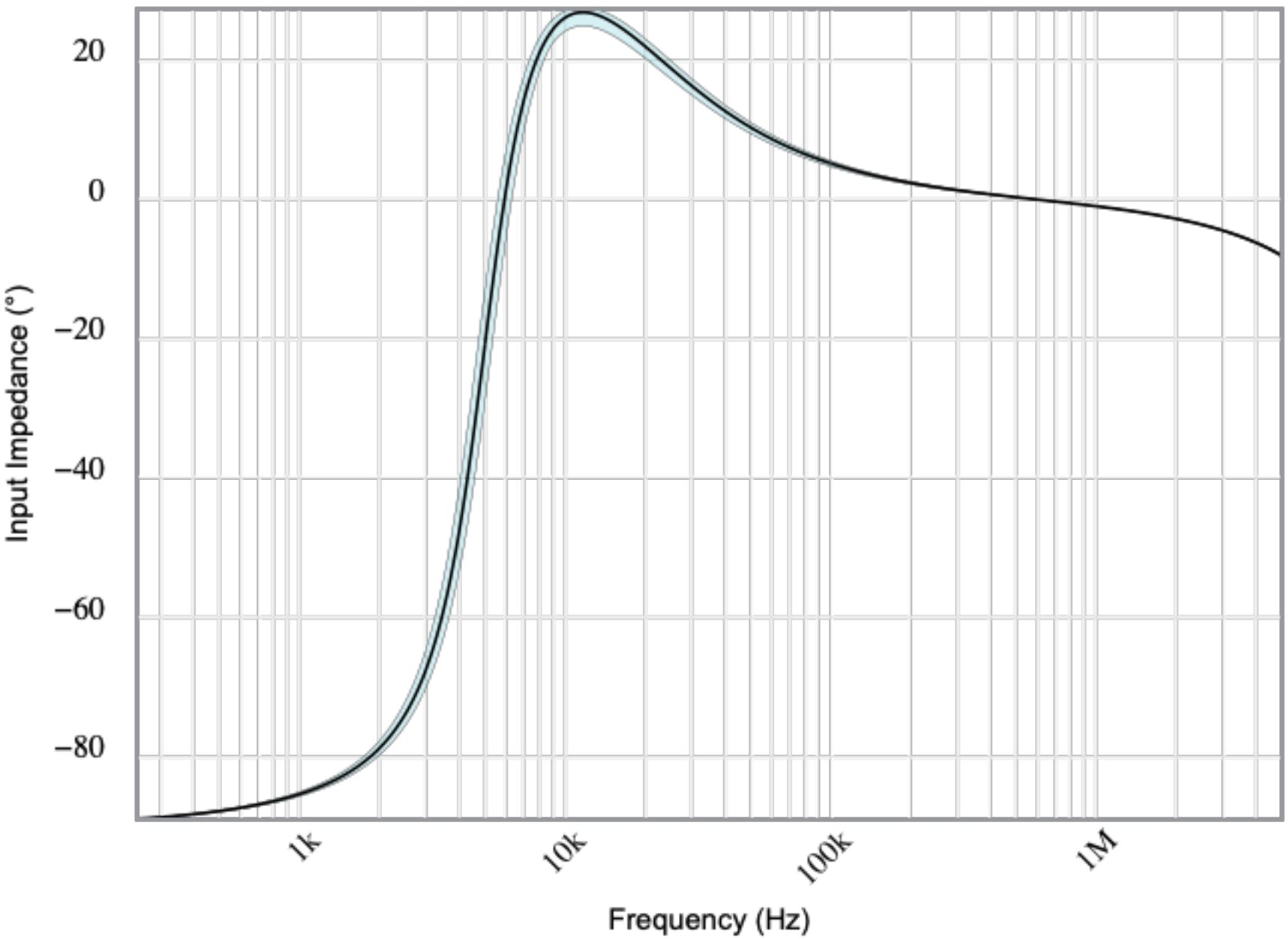
Step Response



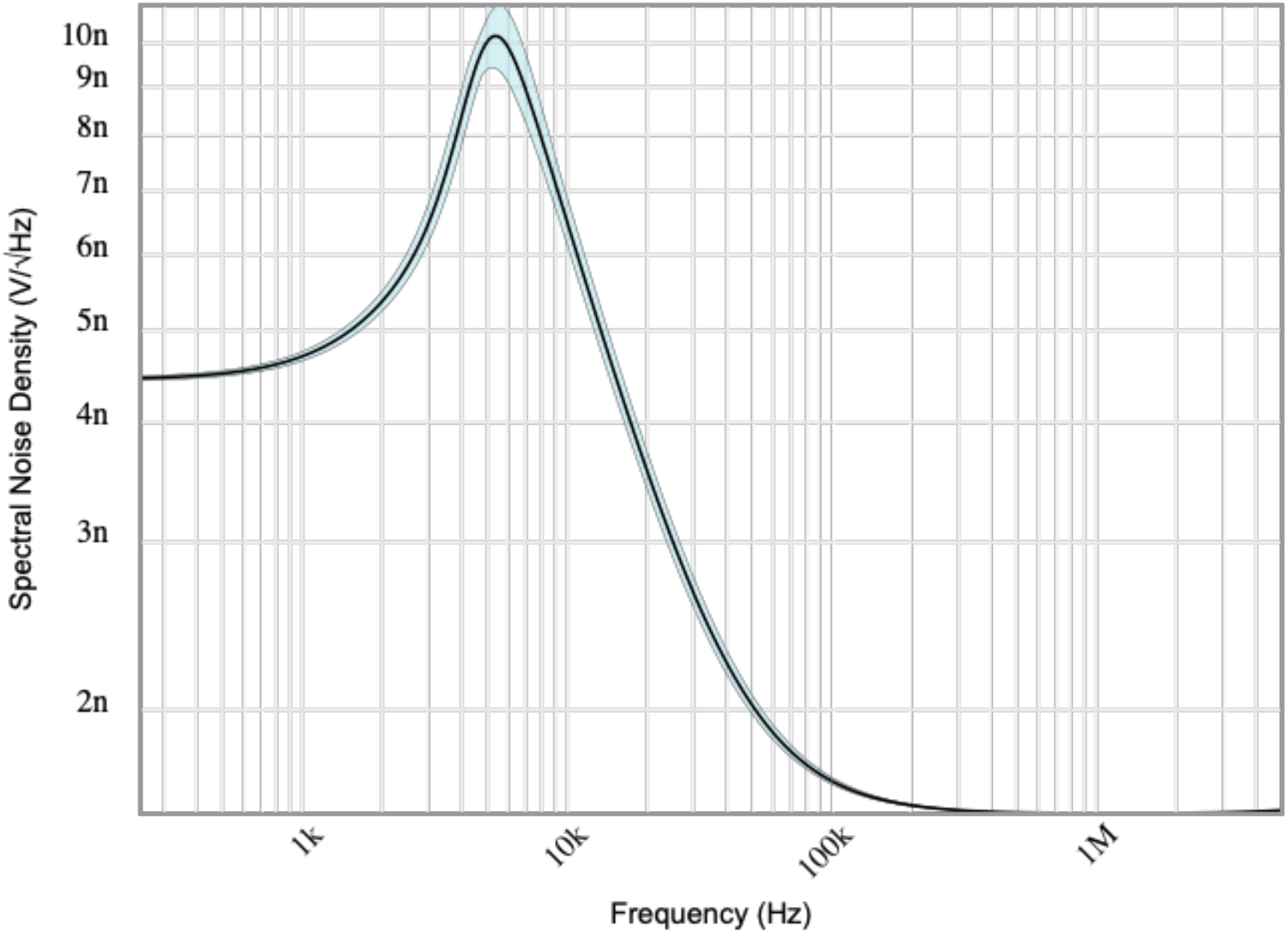
Input Impedance Magnitude



Input Impedance Phase



Noise


10n
9n
8n
7n
6n
5n
4n
3n
2n
Spectral Noise Density (V/√Hz)
1k
10k
100k
1M
Frequency (Hz)

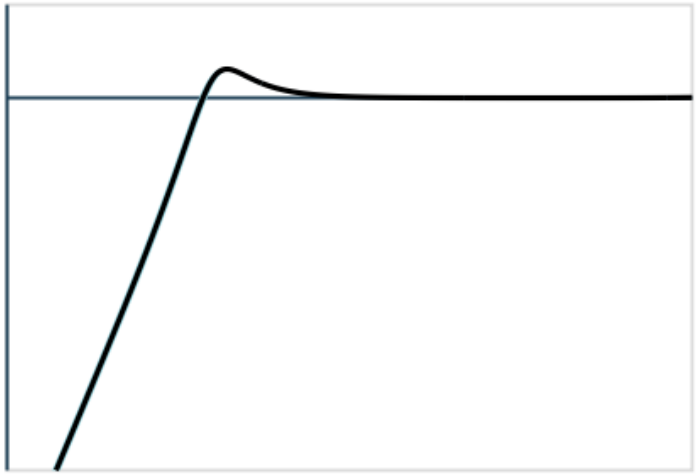
Stages

Your filter requires 2 op amp stage(s) with the following characteristics

A

2nd order High-Pass Sallen Key

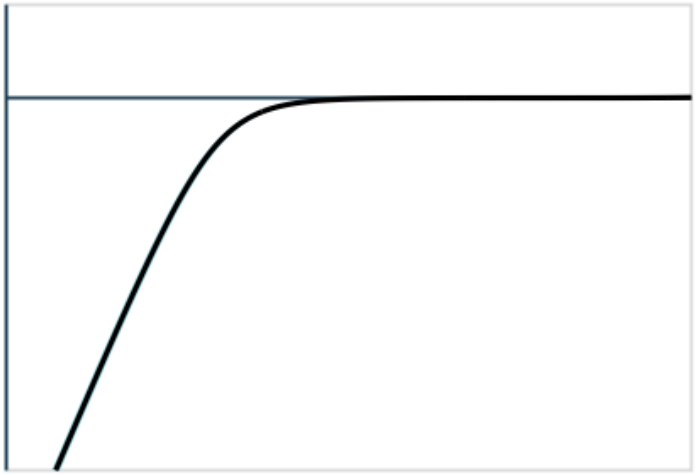
	Target	Simulated
Gain (V/V):	1	1 to 1
f_p (Hz):	5k	4.71k to 5.31k
Q:	1.31	1.31 to 1.33



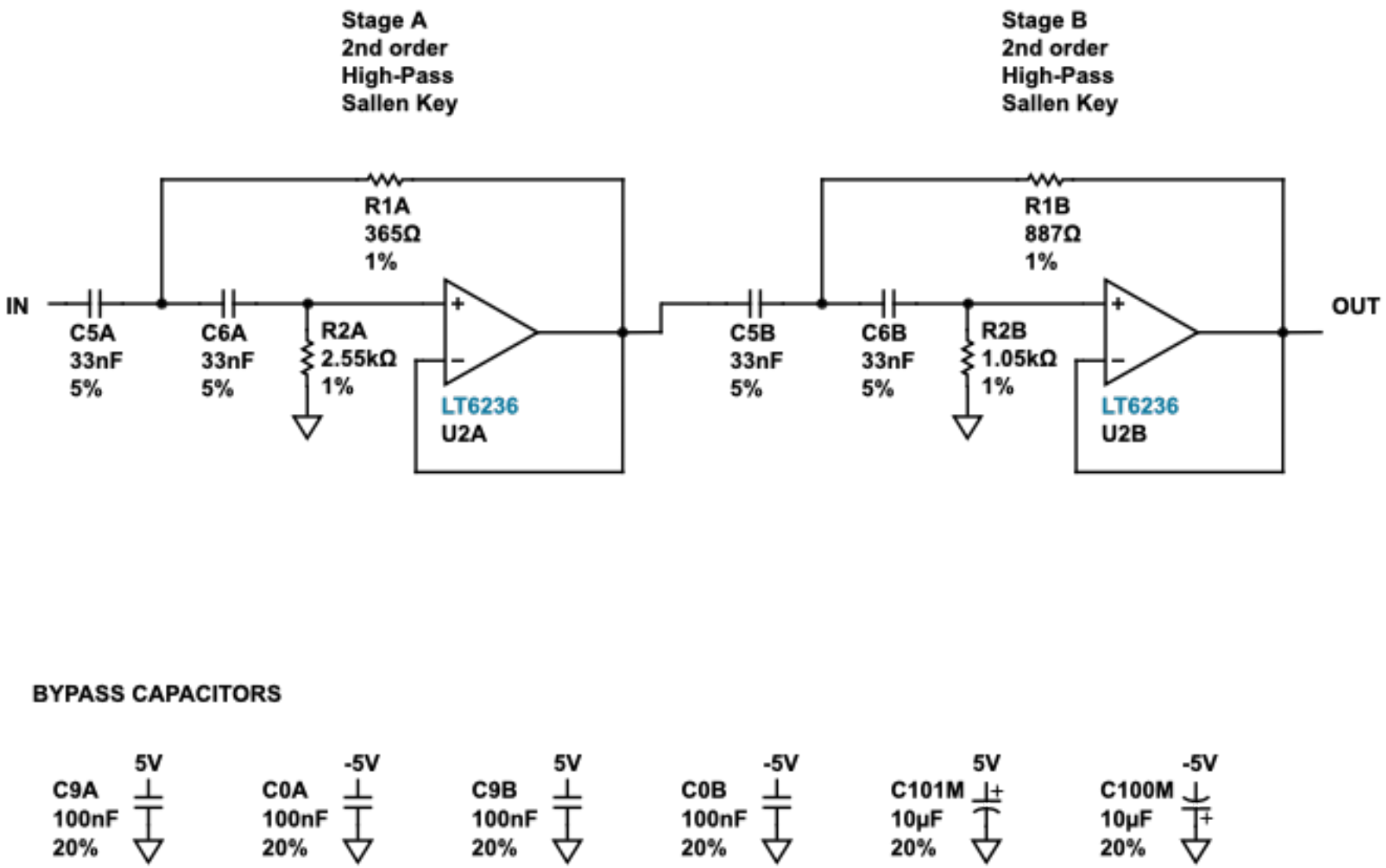
B

2nd order High-Pass Sallen Key

Target	Simulated
1	1 to 1
5k	4.71k to 5.31k
541m	537m to 549m



Circuit


Stage A
2nd order
High-Pass
Sallen Key
Stage B
2nd order
High-Pass
Sallen Key
R1A
365Ω
1%
R1B
887Ω
1%
IN
OUT
C5A
33nF
5%
C6A
33nF
5%
R2A
2.55kΩ
1%
LT6236
U2A
C5B
33nF
5%
C6B
33nF
5%
R2B
1.05kΩ
1%
LT6236
U2B
BYPASS CAPACITORS
C9A
100nF
20%
5V
C0A
100nF
20%
-5V
C9B
100nF
20%
5V
C0B
100nF
20%
-5V
C101M
10µF
20%
5V
C100M
10µF
20%
-5V